



A framework for Measuring sustainability factors in Oman's healthcare system

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ملخص البحث :

يهدف البحث إلى: تحديد عوامل الاستدامة في قطاع الرعاية الصحية ، وقياس عوامل الاستدامة في قطاع الرعاية الصحية في عمان ، وإعطاء الأولوية لأهمية عوامل الاستدامة في قطاع الرعاية الصحية. وقد اشتملت عينة الدراسة على 20 خبير رعاية صحية يتمتعون بخبرة لا تقل عن 10 سنوات في إدارة الرعاية الصحية في 5 مستشفيات عامة مختلفة في مسقط. من المتوقع أن يخرج البحث بالتوصيات المناسبة في ضوء نتائج الدراسة.

الكلمات المفتاحية:

الاستدامة، القطاع الصحي، التحليل الهرمي، الجودة

Measuring sustainability factors in Oman's healthcare

Abstract

The research aims to: Identify factors of sustainability in the healthcare sector, measure sustainability factors in Oman's healthcare sector, and prioritize the importance of sustainability factors in the healthcare sector. The study sample included 20 healthcare experts with at least 10 years of experience in health care management in 5 different public hospitals in Muscat. The research is expected to come up with appropriate recommendations in light of the study.

Keywords: sustainability, healthcare, AHP, quality.

Introduction:

The first decade of the twenty-first century has witnessed strong growth in different sectors, especially in the economy, technology, and healthcare, which create a massive change. Therefore, governments must have the ability to evolve and keep pace with world requirements. If we bring light into the healthcare sector, we will observe that it goes simultaneously with other sectors and it also reflects developments in the overall systems.

The healthcare sector requires significant resources to survive and meet health needs. However, there are challenges that the health sector has faced in providing specialists in multidisciplinary and interdisciplinary healthcare-related processes. Furthermore, there are worries of the wellbeing of the health workers from one side, due to the high burnout levels (Gola et al., 2020), and the health of other organisms from the other side, due to the healthcare pollutants such as medical wastes that are considered to be the main cause of diseases around the health institution (Molero et al., 2020).

Health development plans have been showing remarkable changes in health care in Oman both in distribution and quality. In Oman, there were two hospital units under the ministry of health (MOH) in 1970. The number later increased to 49 in 2000 and it grew further to reach 51 hospital units in 2020. (Ministry of Health, 2021). According to the National Center for Statistics & Information (2021), clinical units under the private sector in 2000 were 560, and then the number increased up to 1230 in 2020. With this increase, the number of doctors per 10,000 of the total population also increased from 0.2 in 1970 to 19.7 in 2020. From a financial side, MOH shared from the total governmental expenditure 6.2% in 2000 with expenditure reaches 145.7 million Omani rials (OMR). The expenditure was then reduced to 4.7% from 2000 to 2010 followed by a dramatic increase to 7.5% in 2020 with expenditure reaching 988.2 million OMR

under the effect of Covid-19 pandemic. In the same period, MOH investment expenditure in 2000 reached 1200 million OMR and then increased to 2210 million OMR only in 2020. As a result, all these statistics indicate that the government must provide more resources in this sector which will require adopting an approach that keeps the system balanced.

Sustainability can be considered a domain of quality in healthcare, extending the responsibility of health services to patients not just of today but of the future. The long term perspective highlights the impacts of our healthcare system on our environment and communities and in turn back into population health. A sustainable approach will therefore expand the healthcare definition of value to measure health outcomes against environmental and social impacts alongside financial costs. Considering sustainability from a clinical perspective means allocating resources appropriately (both human and material) and considering the health and wellbeing of staff – arguably the single most important entity in the sustainable delivery of healthcare.

Yet, sustainability is a complex and challenging topic to measure (Bell and Morse, 2003). Despite that, several studies have attempted to develop methods for its assessment covering different dimensions of sustainability (Nilashi, M. 2019). But when it comes to measuring sustainability in healthcare, a scarce number of researchers have investigated that. As an effort to manage the complexity of measuring healthcare sustainability, Aljaberi et al. used the analytical hierarchical process (AHP). They introduced a new framework for sustainability factors (AlJaberi et al., 2017). Another study examined a new comprehensive approach to measure healthcare sustainability effects through applying it to diabetic foot prevention (Hellstrand & Tang, 2019).

In this study, the researcher will seek to identify and measure the sustainability factors in Oman's healthcare sector and prioritize the importance of sustainability factors.

Literature review:

Oman, over the past three decades, has achieved great improvements in its health care system including an increase in the quality of health services that resulted from reducing the death rate of children and increasing the rate of overall life expectancy to 13 years longer than in the European countries (Al Dhawi et al., 2007). However, there are several challenges that contribute to the sustainability of the healthcare system, which fall into four main influences, including the enabling environment, financial, institutional and demand sustainability (Al Dhawi et al., 2007). Under the supply challenges, there are limitations of resources including the funding, personnel and capital infrastructure, and on the demand part, there is an increase in the patient expectations and an increase in the cost of many treatments due to the improvement in the techniques used (Al Dhawi et al., 2007).

Lennox et al. (2020), reviewed 68 articles demonstrating the sustainability approach application in practice, showing a rise in the application of sustainability approaches in a variety of healthcare settings, including primary, secondary, tertiary, and community healthcare. Due to the vague impact of sustainability approaches, only a few articles reported the outcomes. The gaps emphasize the need for more reports of sustainability approaches in the research field.

Buffoli et al. (2013), established an evaluation approach that considers all three pillars of sustainability, examining social, environmental, and economic viability using a set of criteria and measurable indicators. This research categorized existing operative hospitals and new hospital designs. The authors indicated that the system highlights both global and specific essential areas of sustainability and feasible strategic options to improve sustainability. Technical analysis and qualitative questionnaires used to conduct the evaluation allow for a quantitative assessment of sustainability using a sound scoring mechanism. This study has been thoroughly tested within two hospitals, assessing both an operative structure and the designing phase of a new hospital.

Aljaberi et al. (2017), identified seven sustainability factors in the healthcare industry. Then, described a framework to measure them and eventually prioritized them. Analytical hierarchical process (AHP) was used to manage sustainability complexity and rank 31 sustainability sub-factors based on experienced healthcare specialists' input. The data were collected from five

large hospitals in Abu Dhabi, UAE. This study designed a new framework and presented how important the social dimension of sustainability is. However, the two other sustainability dimensions were not thoroughly investigated in the analysis which might affect the comprehensiveness of this framework. Also, it was limited in terms of geographical area to the UAE.

Hellstrand and Tang (2019), attempted to quantify the sustainability effects of healthcare using a new approach. This was applied to the prevention of diabetic foot complications. They identified three relationships between ecosystems and human health and healthcare systems. They also managed to combine all three dimensions of sustainability within their model. They used the toolbox, which was first introduced by Hellstrand (2015), and its instruments to quantify the performance of sustainability systems. They found that many of the 17 sustainable development goals by the United Nations can be calculated and contribute to preventing diabetic complications.

Since, till now, there are few studies on how sustainability is measured in healthcare and none in Oman, the present study aims to investigate that. It will include more economic and environmental factors of sustainability along with the social factors, that were thoroughly examined by Aljaberi et al. (2017), to come up with a more inclusive framework. Not only that, but it will rank these factors according to the Omani healthcare needs. Thus, it will guide Oman's health care leaders toward building a more sustainable system by focusing on the priorities. Eventually, this is expected to contribute to Oman's vision 2040 by taking part in achieving the objectives of the second national priority; Health (Implementation Follow-up Unit, 2020).

Study Problem:

In the context of the strategic direction of Oman vision 2040, the government seeks to provide the healthcare system with International Standards, which aims to decentralize the healthcare

system operating with quality and diversified and sustainable funding sources for the healthcare system.

The government would typically face many difficulties that have been hampering the promotion of development such as lack of resources in the healthcare system, as can be seen in the paper of Al Dhawi et al. (2007). Moreover, the annual health report 2020 shows a higher expenditure rate of the Ministry of Health compared to the previous three years (2017-2019) (Ministry of Health, 2021). The report indicates that the total recurrent expenditures in MOH reached approximately RO 972.5 million in 2020, compared to approximately RO 793.3 million in 2019, representing a 22.6 percent increase (Ministry of Health, 2021). Therefore, these statistics of higher expenditures in MOH and the lack of resources stress the risk to the long term sustainability.

According to Lennox et al. (2020), insufficient studies are conducted on sustainability in healthcare. Moreover, the authors emphasize the need for more reports in this field which reflects how this approach is important. Furthermore, there is a heavy pressure from the public opinion regarding the development of healthcare that will lead to having an expeditionary reform as well as sustainable resources.

Central Bank of Oman (2020) indicates that, over the last few years, the government has been suffering from a large budget deficit, and it has relied heavily on borrowing to finance the majority of its fiscal deficit. As a result, the government debt has steadily increased to RO 18.9 billion at the end of 2020, up from RO 16.8 billion a year ago, which will significantly drag down growth in the government's performance. Accordingly, assisting the prioritization of sustainability factors would reduce government overheads.

Importance of the study:

- 1-The study will contribute to supporting the 2040 Oman vision.
- 2-The results of the study can benefit the patient and the health care provider by providing health sector spending in attracting more advanced and effective health practices.

3- The study will enable decision-makers to make better judgments, planning and implementation towards sustainable development goals and targets

4-The results of this study will contribute to determining the prioritizing factors to improve the healthcare sector.

5- The framework of measuring sustainability in healthcare could be generalizable to the other service sectors.

5- The study will provide opportunities for researchers to expand the scope of this field by building on the outcomes to enhance the healthcare sector.

Objectives of the study:

1. To identify factors of sustainability in the healthcare sector.
2. To measure sustainability factors in Oman's healthcare sector.
3. To prioritize the importance of sustainability factors in Oman's healthcare sector.

Study questions:

- 1- What are the factors of sustainability in the healthcare sector?
- 2- How to measure sustainability factors in Oman's healthcare sector?
- 3- What are the most important prioritized sustainability factors in Oman's healthcare sector?

Study concepts:

Sustainability

According to the Brundtland report of the World Commission on Environment and Development, sustainable development is defined as "meets the needs of the present without jeopardizing future generations' ability to meet their own needs" (Holdgate, 1987).

In its most basic form, the Oxford English Dictionary (OED) defines sustainability as follows: "The property of being environmentally sustainable; the extent to which a process or enterprise can be maintained or continued while avoiding long-term depletion of natural resources." (Waite, 2012).

Brown et al. (1987) found that the meaning of the word sustainability "is highly dependent on the context in which it is used, as well as whether it is used from a social, economic, or ecological perspective."

Healthcare service: is an intangible product and cannot physically be touched, felt, viewed, counted, or measured like manufactured goods. Producing tangible goods allows quantitative measures since they can be sampled and tested for quality throughout production and later.

Healthcare: It is an effort made by skilled and certified professionals to maintain or restore physical, mental, or emotional well-being.

Methodological procedures:

Study Design:

This is a semi-quantitative cross-sectional study that will be conducted among 5 large public tertiary hospitals in Muscat Governorate, Oman. Data will be collected from twenty healthcare experts using a 9-point scale questionnaire. The participants will have at least ten years of experience in healthcare management. The questionnaire questions are based on a pair-wise comparison in which they will only need to compare two factors at a time. AHP will be used to analyse the answers and compare between the two factors at a time. That also will eventually give us an overall ranking for all factors which will assist the decision-making process for

sustainability solutions. Instead of surveys, we are gathering experts' opinions. Thus, having this small sample size is acceptable when using AHP (Cheng & li, 2001; Saaty, 1990).

The limits of the study:

Spatial limits: Oman

Time limits: Academic year (2022)

Human limits: Workers in the health sector.

Study community: Oman

The study sample: 20 healthcare experts with at least 10 years of experience in health care management in 5 different public hospitals in Muscat.

Study tools:

Analytical hierarchical process model: AHP is a method for ranking decision alternatives and selecting the best one given multiple criteria (Bayazit, 2005). Decision-makers would typically decide based on how the alternatives compare, according to several criteria. selected AHP because it is one of the oldest and most trusted decision-making approaches, and it has been utilized by many researchers to make their decisions.

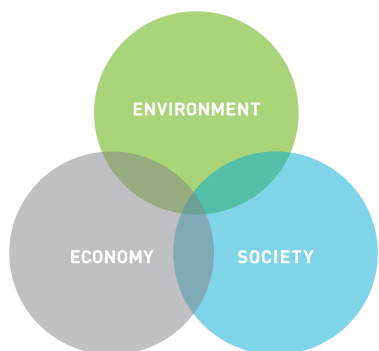
A 9-point scale questionnaire will be designed and distributed to the respondents (managers, specialists and researchers.) to collect their opinions.

Theoretical framework:

Sustainability definition

The idea of "sustainability" is broad, with its essential element being the capacity to "meet present demands without jeopardizing future requirements." It is frequently used in healthcare to refer to a health system's financial sustainability. Financial sustainability in the healthcare sector has been a long-needed issue, and it is frequently a major cause of worry. However, as a larger sustainable policy agenda emerges, the term "sustainability" is evolving to describe the reconciliation of financial, social, and environmental forces that impact current resource usage while not dramatically diminishing what is available for future generations.

Three pillars of sustainability



Environmental sustainability is when ecological integrity is preserved, and all of the earth's environmental systems are kept in balance while the natural resources within them are utilized at a rate that allows them to regenerate themselves.

Economic sustainability is when human communities all over the world are able to keep their independence and have access to the resources (both financial and non-financial) that they

require to satisfy their requirements. Economic systems remain intact, and activities such as secure sources of livelihood are available to everyone.

Social sustainability is when universal human rights and basic requirements are accessible to all individuals who have the means to keep their families and communities healthy and secure. Just leaders guarantee that personal, labor, and cultural rights are recognized and preserved in healthy communities.

According to Merriam Webster dictionary, **health care** means the effort made to maintain or restore physical, mental, or emotional well-being. It is done by well-trained medical professionals and other health staff such as nurse, dentistry, psychologist, and others. (“healthcare”,n.d.)

Health system is all of the mechanisms, resources and processes that together meet the needs of the population. It does not only ease health care for the patient but also prevents healthy people from getting sick.

Health care centers are divided into primary, secondary, tertiary, and quaternary health care centres. The majority of health care centres in Oman is under the ministry of health MOH, and they are classified broadly into two :

1-Hospitals: governorate hospital(secondary and tertiary in some governorate), wilayat hospital, and local hospital.

2-health centres: health center, health center with beds, and extended health centers.

The responsibility of ensuring the availability of healthcare in Oman is on The Ministry Of Health (MOH)(MOH, 2020).

Timeline and Milestones

Period	Tasks
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1 month	- Literature review - Identifying the main factors to measure sustainability and subfactors.
2 months	- Developing a 9-point scale questionnaire with a pairwise comparison - Validating the questionnaire
4 months	- Selecting the healthcare experts from 5 different tertiary hospitals in Muscat - Setting appointments for the interviews - Collecting data through answering the questionnaire
1.5 month	- Data analysis

2.5 months	- Thesis write up
1 month	- Working on publishing process
12 months	total

References

- al Dhawi, A. A., West Jr, D. J., Spinelli, R. J., & Gompf, T. A. (2007). The Challenge of Sustaining Healthcare in Oman. In *The Health Care Manager* (Vol. 26, Issue 1).
- AlJaberi, O. A., Hussain, M., & Drake, P. R. (2017). A framework for measuring sustainability in healthcare systems. *International Journal of Healthcare Management*, 13(4), 276–285. <https://doi.org/10.1080/20479700.2017.1404710>
- Bayazit, O. (2005). Use of AHP in decision-making for flexible manufacturing systems. *Journal of Manufacturing Technology Management*, 16(7), 808–819. <https://doi.org/10.1108/17410380510626204>
- Bell, S., & Morse, S. (2013). *Measuring sustainability: Learning from doing*. Routledge.
- Brown, B. J., Hanson, M. E., Liverman, D. M., & Merideth, R. W. (1987). Global sustainability: Toward definition. *Environmental Management*, 11(6), 713–719. <https://doi.org/10.1007/BF01867238>
- Buffoli, M., Capolongo, S., Bottero, M., Cavagliato, E., Speranza, S., & Volpatti, L. (2013). Sustainable healthcare: how to assess and improve healthcare structures' sustainability. *Annali Di Igiene : Medicina Preventiva e Di Comunità*, 25(5), 411–418. <https://doi.org/10.7416/ai.2013.1942>
- Central Bank of Oman. (2021). *Annual Report 2020*.
- Cheng, E. W. I., & li, H. (2001). Analytic hierarchy process: An approach to determine measures for business performance. In *Measuring Business Excellence* (Vol. 5, Issue 3, pp. 30–37). <https://doi.org/10.1108/EUM00000000005864>

- Gola, M., Brambilla, A., Barach, P., Signorelli, C., & Capolongo, S. (2020). Educational Challenges in Healthcare Design: Training Multidisciplinary Professionals for Future Hospitals and Healthcare. *Annali Di Igiene Medicina Preventiva e Di Comunita*, 32(5), 549–566. <https://doi.org/10.7416/ai.2020.2375>
- Hellstrand, S., & Tang, U. (2019). *The role of apps in diabetic foot education View project*. <https://doi.org/10.3827/faoj.2019.1203.0005>
- Hellstrand, Stefan. (2015). *On the value of land*. Mälardalen University.
- Holdgate, M. W. (1987). Our Common Future: The Report of the World Commission on Environment and Development. Oxford University Press, Oxford & New York: xv+ 347+ 35 pp., 20.25× 13.25× 1.75 cm, Oxford Paperback,£ 5.95 net in UK, 1987. *Environmental Conservation*, 14(3), 282.
- Implementation Follow-up Unit. (2020). *Oman Vision 2040*.
- Lennox, L., Linwood-Amor, A., Maher, L., & Reed, J. (2020). Making change last? Exploring the value of sustainability approaches in healthcare: a scoping review. *Health Research Policy and Systems*, 18(1). <https://doi.org/10.1186/s12961-020-00601-0>
- Ministry of Health. (2021). *Annual Health Report 2020*.
- Molero, A., Calabrò, M., Vignes, M., Gouget, B., & Gruson, D. (2020). Sustainability in healthcare: Perspectives and reflections regarding laboratory medicine. In *Annals of Laboratory Medicine* (Vol. 41, Issue 2, pp. 139–144). Seoul National University, Institute for Cognitive Science. <https://doi.org/10.3343/alm.2021.41.2.139>
- Nilashi, M., Rupani, P. F., Rupani, M. M., Kamyab, H., Shao, W., Ahmadi, H., ... & Aljojo, N. (2019). Measuring sustainability through ecological sustainability and human sustainability: A machine learning approach. *Journal of Cleaner Production*, 240, 118162.
- Saaty, T. L. (1990). *Decision making for leaders: the analytic hierarchy process for decisions in a complex world*. RWS publications.
- Waite, M. (2012). *Paperback Oxford English dictionary*. Oxford University Press.
- National center for information and statistics (.n.d.).retrieved from <https://data.gov.om/suirlic/health>

- healthcare. 2022. In *Merriam-Webster.com*. Retrieved January 27, 2022, from <https://www.merriam-webster.com/dictionary/health%20care>.